Digital CAD Training and Services

SolidWorks 40 Days Syllabus

Day-wise learning plan with daily practice tasks

Duration	Software	Training Focus
40 Days	**SolidWorks**	Sketching, Part Modeling, Surface, Assembly, Drafting

A complete, structured training document for students to understand the learning path, daily topics and practice expectations before starting the course.

Course Objective

This syllabus is designed to build a strong practical foundation in SolidWorks through a clear 40-day progression. Students start with essential sketching concepts, move into feature-based part modeling, continue with surface creation and quality checks, and finish with assemblies and production drawing preparation.

What Students Will Learn

- Create fully defined sketches using dimensions and relations.
- Build solid models using extrude, revolve, sweep, loft, shell, rib, pattern and mirror tools.
- Create and edit surface features including trims, knit, offsets and surface analysis.
- Prepare assemblies using mates, exploded views and basic component positioning.
- Create manufacturing drawings with standard views, section views, dimensions, tolerances and GD&T.
- Complete daily practice tasks to improve hands-on command and design confidence.

Module Breakdown

Module	Days	Focus Area
Sketch	**5**	2D profile creation, sketch constraints, dimensions and sketch checking
Part Modeling	**15**	Core 3D feature creation, modification and feature management
Surface	**8**	Surface generation, trimming, knitting, offsets and analysis
Assembly	**5**	Component insertion, mates and exploded view creation
Drafting	**7**	Drawing setup, views, dimensions, tolerances, GD&T and final sheet setup

Complete Day-wise Syllabus

Each day includes one focused topic and one daily practice task. Students should complete the practice task on the same day to maintain continuity and improve software command.

Day	Module	Topic	Daily Practice Task
1	Sketch	Interface + Navigation	Create basic plate sketch
2	Sketch	Line / Circle / Arc	Create bracket profile
3	Sketch	Geometric Relations	Apply constraints
4	Sketch	Smart Dimensions	Modify sketch
5	Sketch	Sketch Analysis	Fully constrain complex sketch
6	Part Modeling	Extrude Boss/Base	Create base solid
7	Part Modeling	Revolve	Create cylindrical part
8	Part Modeling	Hole Wizard	Create hole features
9	Part Modeling	Fillet	Apply fillets
10	Part Modeling	Chamfer	Apply chamfers
11	Part Modeling	Draft	Apply draft
12	Part Modeling	Shell	Hollow part
13	Part Modeling	Rib	Create reinforcement
14	Part Modeling	Pattern	Create patterns
15	Part Modeling	Mirror	Mirror features
16	Part Modeling	Sweep	Create sweep feature
17	Part Modeling	Loft	Create loft feature
18	Part Modeling	Boundary Boss	Create boundary feature
19	Part Modeling	Combine	Combine bodies
20	Part Modeling	Move Face	Modify geometry
21	Surface	Extruded Surface	Create surface
22	Surface	Revolved Surface	Create revolve surface
23	Surface	Swept Surface	Create sweep
24	Surface	Lofted Surface	Create loft surface
25	Surface	Trim Surface	Trim surfaces
26	Surface	Knit Surface	Join surfaces
27	Surface	Offset Surface	Create offset
28	Surface	Surface Analysis	Check quality
29	Assembly	Assembly Overview	Insert parts
30	Assembly	Mate - Coincident	Apply constraint
31	Assembly	Mate - Distance	Apply offset
32	Assembly	Mate - Angle	Apply angle
33	Assembly	Exploded View	Create explode
34	Drafting	Drawing Setup	Create drawing

Day	Module	Topic	Daily Practice Task
35	Drafting	Standard Views	Generate views
36	Drafting	Section View	Create section
37	Drafting	Dimensioning	Add dimensions
38	Drafting	Tolerances	Apply tolerances
39	Drafting	GD&T	Apply GD&T
40	Drafting	Sheet Setup	Prepare final sheet

Student Practice Guidelines

- Complete the daily practice task before moving to the next day.
- Keep every file properly named with day number, topic and your name.
- Maintain a separate practice folder for Sketch, Part Modeling, Surface, Assembly and Drafting.
- Revise previous commands weekly to avoid tool confusion and improve speed.
- Submit doubts with screenshots, model file name and the exact step where you are stuck.

Final Outcome After 40 Days

Area	Expected Outcome
Sketching	Students can create and modify constrained sketch profiles.
Part Modeling	Students can build practical 3D parts with common production features.
Surface	Students can create, trim, knit and analyze basic surface geometry.
Assembly	Students can insert components, apply mates and create exploded views.
Drafting	Students can prepare drawings with views, dimensions, tolerances and GD&T.

For course details and batch updates, contact Digital CAD Training and Services.